

Summary of Changes from BSIM-SOI 100.1.0 to BSIM-SOI 100.1.1

BSIM Group, IIT Kanpur, UC Berkeley

List of authors: https://www.bsim.berkeley.edu/models/bsimsoi/auth_soi_s/

BSIMSOI 100.1.1_beta1

A. Summary of bug fixes:

1. **2023bug17 (GF)**: Removal of the kink in the I_{ds} vs. V_g at different V_e for CDSBS1=0.1
2. **2023bug18 (Cadence)**: Correcting TNODEOUT dependency of AGBCP2 region gate to body tunnelling current.
3. **2023bug19 (Cadence)**: Removal of redundant assignment of V_{gfb} , v_{gfb1} .
4. **2023bug20 (Cadence)**: Adding g_{min} between $bi2$ and si/di nodes.
5. **2023bug21 (GF)**: Correcting dv_{th_dibl} definition for SOIMOD=1.
6. **2023bug22 (GF/Cadence)**: Correcting overflow issue related to mobility model.
7. **2023bug23 (Cadence)**: Correcting the definition of BODYMOD mod-selectors and addition of warnings.
8. **2024bug1 (GF)**: Correcting the length and width dependencies of V_{FB_i} , V_{FBCV_i} , V_{SATCV_i} .
9. **2024bug2 (Cadence)**: Removal of Floating-point errors.
10. **2024bug3 (Synopsys)**: Correcting the reference node voltage definition for calculation of Gate current AGBCP2 FET.
11. **2024bug4 (Synopsys)**: Correcting Floating body definition in 5 terminal devices with TNODEOUT=1.
12. **2024bug5 (ADI)**: Additional tests are added to the QA package.
13. **2024bug6 (ADI)**: Removal of VAMPyRE warnings/errors.
14. **2024bug7 (UCB)**: Correcting the definition of VFBB and updating the Technical Manual.

BSIMSOI 100.1.1_beta2

B. Summary of bug fixes:

15. **2025bug1 (ADI)**: Removal of VAMPyRE warnings.
16. **2025bug2 (UCB/IITK)**: Updating the Technical Manual.
17. **2025bug3 (UCB/IITK)**: Removal of unused “tempc1” variable from code.

A. Description of bug-fixes:

1. 2023bug17 (GF): Removal of the kink in the I_{ds} vs. V_g at different V_e for CDSBS1=0.1

- The kink in I_{ds} vs. V_g for different V_e (substrate bias) at CDSBS1=0.1 has been removed.
- Following expressions are updated in this regard:

BSIM-SOI 100.1.0

```
phic_s = PHIC-RC*(vgfbb*nVt) + (1.0+RC)*(phisb0);
```

BSIM-SOI 100.1.1_beta1

```
phic_s = PHIC-RC*(vgfbb*nVt)*CDSBS1_i+ (1.0+RC)*(phisb0);
```

2. 2023bug18 (Cadence): Correcting TNODEOUT dependency of AGBCP2 region gate to body tunnelling current.

- The TNODEOUT dependency of AGBCP2 region gate to body tunnelling current is updated.
- The corrections are shown below:

BSIM-SOI 100.1.0

```
.....//Gate tunneling component in the AGBCP2 region
if ($port_connected(b) != 0 && AGBCP2 > 0) begin
```

BSIM-SOI 100.1.1_beta1

```
.....//Gate tunneling component in the AGBCP2 region
if ((BODYMOD != 0 && $port_connected(b) != 0 && !(TNODEOUT && !$port_connected(t)) && MODAGBCP2 == 1) && AGBCP2 > 0) begin
```

3. 2023bug19 (Cadence): Removal of redundant assignment of Vgfb, vgb1.

- The following redundant assignment of Vgfb and vgb1 have been removed.

BSIM-SOI 100.1.0

```
.....//Normalized terminal voltages for DDMOD
vgfb1 = vgbAGBCP2;
vgfbb = vel - vfb;
```

BSIM-SOI 100.1.1_beta1

Removed!

4. 2023bug20 (Cadence): Adding gmin between bi2 and si/di nodes.

- The gmin is added between the bi2 and si/di internal nodes.
- The changes in this regard are as follows:

BSIM-SOI 100.1.1_beta1

```
if ( (BODYMOD !=0) && ($port_connected (b) == 1) && (!(TNODEOUT == 1)
&& ($port_connected(t) == 0))) && (MODAGBCP2 == 1) ) begin
    I(bi2, si) <+ V(bi2, si) * gmin;
    I(bi2, di) <+ V(bi2, di) * gmin;
end
```

5. 2023bug21 (GF): Correcting dvth_dibl definition for SOIMOD=1.

- The **dvth_dibl** term has never been used and hence removed.
- The parameters ETAE, ETA1, ETAEDGE and ETA1EDGE are removed.
- The changes in this regard are as follows:

BSIM-SOI 100.1.0

```
if (SOIMOD == 1)begin
    dvth_dibl = -(ETA0_t + ETAB_i * Vbsx + ETAE_i * Vesx) *
    Theta_DIBL * (Vdsx + ETA1_i * sqrt(Vdsx + 0.01));
end
```

BSIM-SOI 100.1.1_beta1

Removed for both main and EDGEFET!

6. 2023bug22 (GF/Cadence): Correcting overflow issue related to mobility model.

- Solved using correcting FPE issues as discussed below in 2024bug2.

7. 2023bug23 (Cadence): Correcting the definition of BODYMOD mod-selectors and addition of warnings.

- The definition of **BODYMOD** selector is updated with added warnings.
- The following is updated in this regard:

BSIM-SOI 100.1.0

```
.....//Body.Contact.resistance
.....RBODYEXT.=.NRB.*.RBSH;
.....if.($port_connected(b).==.0).begin
.....V(b,.bi).<+.0;
.....V(b,.bi2).<+.0;
```

BSIM-SOI 100.1.1_beta1

```
.....//Body.Contact.resistance
.....RBODYEXT.=.NRB.*.RBSH;
.....if.((BODYMOD==.0)||.($port_connected(b).==.0)).begin
.....V(b,.bi).<+.0;
.....V(b,.bi2).<+.0;
```

Following warnings have been added:

```
].....if.($port_connected(b).==.0).begin
].....if.(BODYMOD!=.0).begin
].....$strobe("\nWarning: b-node is not connected, however BODYMOD!=.0. Assuming Floating-body device!");
].....end
].....if.(MODAGBCP2!=.0).begin
].....$strobe("\nWarning: MODAGBCP2 should be set to 0 as it is a floating-body device!");
].....end
].....end.else.begin
].....if.(TNODEOUT.&&.!$port_connected(t)).begin
].....if.(BODYMOD!=.0).begin
].....$strobe("\nWarning: last node is b which is a thermal node, however BODYMOD!=.0. Assuming Floating-body device!");
].....end
].....if.(MODAGBCP2!=.0).begin
].....$strobe("\nWarning: MODAGBCP2 should be set to 0 as it is a floating-body device!");
].....end
].....end.else.begin
].....if.(BODYMOD==.0).begin
].....$strobe("\nWarning: b-node is connected, however BODYMOD=.0. Assuming Floating-body device!");
].....if.(MODAGBCP2!=.0).begin
].....$strobe("\nWarning: MODAGBCP2 should be set to 0 as it is a floating-body device!");
].....end
].....end
].....end
].....end
].....if.(. (BODYMOD!=.0) && ($port_connected(b).==.1) && !((TNODEOUT==.1) && ($port_connected(t).==.0))) .begin
].....if.(MODAGBCP2==.0).begin
].....$strobe("\nWarning: MODAGBCP2 should be set to 1 as it is a body-contacted device!");
].....end
].....end
```

8. 2024bug1 (GF): Correcting the length and width dependencies of VFB_i, VFBCV_i, VSATCV_i.

- The definition of VFB_i, VFBCV_i, VSATCV_i are corrected in terms of the following:

BSIM-SOI 100.1.0

```
T0      = VFBL * max(pow(Inv_Lact, VFBLEXP) - pow(Inv_Llong,
VFBLEXP), 0.0);
T1      = VFBW * max(pow(Inv_Wact, VFBWEXP) - pow(Inv_Wwide,
VFBWEXP), 0.0) + VFBWL * pow(Inv_WL, VFBWLEXP);
VFB_i   = VFB_i * (1.0 + T0 + T1);
T0      = VFBCVL * max(pow(Inv_Lact, VFBCVLEXP) - pow(Inv_Llong,
VFBCVLEXP), 0.0);
T1      = VFBCVW * max(pow(Inv_Wact, VFBCVWEXP) - pow(Inv_Wwide,
VFBCVWEXP), 0.0) + VFBCVWL * pow(Inv_WL, VFBCVWLEXP);
VFBCV_i = VFBCV_i * (1.0 + T0 + T1);
T0      = VSATCVL * max(pow(Inv_Lact, VSATCVLEXP) - pow(Inv_Llong,
VSATCVLEXP), 0.0);
T1      = VSATCVW * max(pow(Inv_W, VSATCVWEXP) - pow(Inv_Wwide,
VSATCVWEXP), 0.0) + VSATCVWL * pow(Inv_WL, VSATCVWLEXP);
VSATCV_i = VSATCV_i * (1.0 + T0 + T1);
```

BSIM-SOI 100.1.1_beta1

```
T0      = VFBL * max(pow(Inv_L, VFBLEXP) - pow(Inv_Llong, VFBLEXP),
0.0);
T1      = VFBW * max(pow(Inv_W, VFBWEXP) - pow(Inv_Wwide, VFBWEXP),
0.0) + VFBWL * pow(Inv_WL, VFBWLEXP);
VFB_i   = VFB_i * (1.0 + T0 + T1);
T0      = VFBCVL * max(pow(Inv_Lact, VFBCVLEXP) - pow(Inv_Llong,
VFBCVLEXP), 0.0);
T1      = VFBCVW * max(pow(Inv_Wact, VFBCVWEXP) - pow(Inv_Wwide,
VFBCVWEXP), 0.0) + VFBCVWL * pow(Inv_Wact * Inv_Lact, VFBCVWLEXP);
VFBCV_i = VFBCV_i * (1.0 + T0 + T1);
T0      = VSATCVL * max(pow(Inv_Lact, VSATCVLEXP) - pow(Inv_Llong,
VSATCVLEXP), 0.0);
T1      = VSATCVW * max(pow(Inv_Wact, VSATCVWEXP) - pow(Inv_Wwide,
VSATCVWEXP), 0.0) + VSATCVWL * pow(Inv_Wact * Inv_Lact, VSATCVWLEXP);
VSATCV_i = VSATCV_i * (1.0 + T0 + T1);
```

Note: Also, some Vampyre errors have been addressed.

9. 2024bug2 (Cadence): Removal of Floating-point errors.

- The following is updated in this regard both in SOIMOD=0 and SOIMOD=1:

Earlier:

```
SP_S_temp = SP_S_pC * SP_S_pC - 2.0 * (SP_S_qC * SP_S_temp);  
sp_dd      = -SP_S_y0 - 2.0 * (SP_S_qC / (SP_S_pC + sqrt(SP_S_temp)));
```

Proposed:

```
SP_S_temp = SP_S_pC * SP_S_pC - 2.0 * (SP_S_qC * SP_S_temp);  
SP_S_temp = sqrt(SP_S_temp * SP_S_temp + 64e-8) - 8e-4;  
sp_dd      = -SP_S_y0 - 2.0 * (SP_S_qC / (SP_S_pC + sqrt(SP_S_temp)));
```

Earlier: T2 = pow(0.5 * (1.0 + (qis / qbs)), UCS_a);

Proposed: T2 = **lexp**(UCS_a * **lln**(0.5 * (1.0 + (qis / qbs))));

Earlier: T7 = pow((Vds / Vdsat) + 1.0e-6, 1.0 / DELTA_t);

Proposed:

```
if (Vdsat == 0.0) begin  
    T7 = 0.0;  
end else begin  
    T7 = pow((Vds / Vdsat) + 1.0e-6, 1.0 / DELTA_t);  
end
```

Earlier: T2 = pow(0.5 * (1.0 + (qim / qbm)), UCS_a);

Proposed: T2 = **lexp**(UCS_a * **lln**(0.5 * (1.0 + (qim / qbm))));

Earlier: T0 = (qiaCV + ETAQM * (qbaCV)) / QM0;

Proposed:

```
T0 = (qiaCV + ETAQM * (qbaCV)) / QM0;  
T0 = hypsmooth(T0,1.0e-3);
```

Earlier: tempc1 = pC + sqrt(tempc);

Proposed:

```
if (tempc >= 0) begin  
    tempc1 = pC + sqrt(tempc);  
end else begin  
    tempc1 = 0.0;  
end
```

```

Earlier:  sqd = sqrt(Pd);
           //-----
           xgd = gam * sqrt(Pd + Dd);
           qid = gam2 * Dd * nVt / (xgd + gam * sqd);

Proposed:  if (Pd >= 0.0) begin
               sqd = sqrt(Pd);
             end else begin
               sqd = 0.0;
             end
           //-----
           xgd = gam * sqrt(Pd + Dd);

           if (xgd == 0.0 && sqd == 0.0) begin
               qid = 0.0;
             end else begin
               qid = gam2 * Dd * nVt / (xgd + gam * sqd);
             end

Earlier:  sqm = sqrt(Pm);

Proposed:  if(Pm >= 0.0) begin
               sqm = sqrt(Pm);
             end else begin
               sqm = 0.0;
             end

```

Note: Here, ‘**Earlier**’ indicates **BSIMSOI 100.1.0** and ‘**Proposed**’ defines **BSIM-SOI 100.1.1_beta1**

10.2024bug3 (Synopsys): Correcting the reference node voltage definition for calculation of Gate current AGBCP2 FET.

- The following is updated in this regard both in SOIMOD=0 and SOIMOD=1:

BSIM-SOI 100.1.0

```

//Gate tunnelling component in the AGBCP2 region

if ((BODYMOD !=0 && $port_connected(b) != 0 && !(TNODEOUT &&
!$port_connected(t)) && MODAGBCP2 ==1) && AGBCP2 > 0) begin

Vgb = devsign * V(gi, b);

.

.

```

BSIM-SOI 100.1.1_beta1

```

//Gate tunnelling component in the AGBCP2 region

if ((BODYMOD !=0 && $port_connected(b) != 0 && !(TNODEOUT &&
!$port_connected(t)) && MODAGBCP2 ==1) && AGBCP2 > 0) begin

Vgb = devsign * V(gi, bi2);

.

.

```

11. 2024bug4 (Synopsys): Floating body definition in 5 terminal devices with TNODEOUT=1.

- When TNODEOUT=1 and 5th terminal (b) to be treated as THERMAL node, b and bi/bi2 (internal nodes) were shorted to ensure floating-body status. This is making thermal node (b) to be shorted with the internal body nodes (bi, bi2).
- The following update is made in this regard.

BSIM-SOI 100.1.0

```
if ((BODYMOD == 0) || ($port_connected(b) == 0)) begin
    V(b, bi) <+ 0;
    V(b, bi2) <+ 0;
end else begin
    if (TNODEOUT && !$port_connected(t)) begin
        V(b, bi) <+ 0.0;
        V(b, bi2) <+ 0.0;
    end else begin
        .
        .
        .
    end
end
```

BSIM-SOI 100.1.1_beta1

```
if ((BODYMOD == 0) || ($port_connected(b) == 0)) begin
    V(b, bi) <+ 0;
    V(b, bi2) <+ 0;
end else begin
    if (TNODEOUT && !$port_connected(t)) begin
        V(bi, bi2) <+ 0.0;
    end else begin
        .
        .
        .
    end
end
```

Additional Warnings added:

```
if (TNODEOUT && !$port_connected(t)) begin
    if (BODYMOD != 0) begin
        $strobe("\n Warning: last node is b which is a thermal node.
        Assuming Floating body device!");
    end else begin
        $strobe("\n Warning: last node is b which is a thermal node,
        however, set BODYMOD != 0. Assuming Floating body device!");
    end
End
```


12.2024bug5 (ADI): Additional tests are added to the QA package.

BSIM-SOI 100.1.1_beta1

Additional tests have been added to the QA package of BSIM-SOI 100.1.1_beta1 code. Kindly refer to the updated QA suite.

13.2024bug6 (ADI): Removal of VAMPyRE warnings/errors.

Corrections have been made in the code as per the VAMPyRE style errors etc. Extra space at the end of Desc attribute have been removed and extra space has been added between ‘)’ and ‘begin’, wherever needed.

Additionally, following changes have been made to remove the warning of possible division by 0 for parameter VTM00:

BSIM-SOI 100.1.0

```
MPRnb( VTM00.....,0.026.....,"V".....,"Hard coded 25 degC thermal voltage" )
```

BSIM-SOI 100.1.1_beta1

```
MPRnb( VTM00.....,0.026.....,"V".....,0.0,1.....,"Hard coded 25 degC thermal voltage" )
```

14.2024bug7 (UCB): Correcting the definition of VFBB and updating the Technical Manual.

BSIM-SOI 100.1.1_beta1

- Definition of VFBB has been updated to ‘Flatband voltage for back gate’ from ‘Flatband voltage’.
- Technical Manual has been updated with relevant references.
- Release note section of the technical manual has been updated.

B. Description of bug-fixes:

15.2025bug1 (ADI): Removal of VAMPyRE warnings.

- Possible divide by 0 errors has warnings removed.
- The if-else conditions pertaining to VAMPyRE (1.9.6) warnings, have been removed.
- Following expressions are updated in this regard. The red dashed boxes (BSIM-SOI 100.1.1 Beta1) indicate the if-else statements that have been removed by the smoothed variables highlighted in green dashed boxes (BSIM-SOI 100.1.1 Beta2).

BSIM-SOI 100.1.1 Beta1

Related to bias-dependent if-else conditions:

```

4330 ..... Ps ..... =. (vgfb1-x7_s) * (vgfb1-x7_s) * (1.0/gam2) .-. Ds;
4331 ..... if. (Ps.>=.0.0) .begin
4332 ..... sqms ..... =. sqrt(Ps);
4333 ..... end.else.begin
4334 ..... sqms ..... =. 0.0;
i. 4335 ..... end

4349 ..... if. (qbs==.0.0) .begin
4350 ..... T2=.0.5;
4351 ..... end.else.begin
4352 ..... T2=.1exp(UCS_a.*.lln(0.5.*.(1.0+.(qis./qbs)))));
ii. 4353 ..... end

4396 ..... Vdsat ..... =. Vdsat./AbulkIV;
4397 ..... if. (Vdsat==.0.0) .begin
4398 ..... T7=.0.0;
4399 ..... end.else.begin
4400 ..... T7 ..... =. pow((Vds./Vdsat).+.1.0e-6,.1.0./DELTA_t);
iii. 4401 ..... end

4705 ..... Pd ..... =. (vgfb1-x7_d) * (vgfb1-x7_d) * (1.0/gam2) .-. Dd;
4706 ..... if. (Pd.>=.0.0) .begin
4707 ..... sqd ..... =. sqrt(Pd);
4708 ..... end.else.begin
4709 ..... sqd ..... =. 0.0;
iv. 4710 ..... end

4714 ..... if. (xgd==.0.0.&&.sqd==.0.0) .begin
4715 ..... qid=.0.0;
4716 ..... end.else.begin
4717 ..... qid ..... =. gam2.*.Dd.*.nVt./.(xgd+.gam.*.sqd);
v. 4718 ..... end

4727 ..... Pm ..... =. (vgfb1-x_m) * (vgfb1-x_m) * (1.0/gam2) .-. Dm;
4728 ..... xgm ..... =. gam.*.sqrt(Dm+.Pm);
4729 ..... if. (Pm.>=.0.0) .begin
4730 ..... sqm ..... =. sqrt(Pm);
4731 ..... end.else.begin
4732 ..... sqm ..... =. 0.0;
vi. 4733 ..... end

```

```

4750 .....Pm.....=..(.vgfb1-x_m+.u_pd.)*(.vgfb1-x_m+.u_pd)*(1/gam2)-..Dm/.km.;
4751 .....xgm.....=.gam*.sqrt(Dm+.Pm);
4752 .....km0.....=.1.0-.Em+.2.0*(xgm*.eta_p*.inv_gam2);
4753 .....x_ds.....=.x_ds*.km*(d0+.D_bar)/.(km0+.km*.D_bar);
4754 .....dps.....=.x_ds*.nVt;
4755 .....if.(Pm>=.0.0).begin
4756 .....sgm.....=.sqrt(Pm);
4757 .....end.else.begin
4758 .....sgm=.0.0;
vii. 4759 .....end

4763 .....if.(x_ds==.0.0).begin
4764 .....alpha_DD.....=.0.0;
4765 .....end.else.begin
4766 .....alpha_DD.....=(qis-.qid)/.dps;
viii. 4767 .....end

4775 .....if.(qbm==.0.0).begin
4776 .....T2=.0.5;
4777 .....end.else.begin
4778 .....T2.....=lexp(UCS_a*.lln(0.5*(1.0+. (qim/.qbm))));
ix. 4779 .....end

4952 .....qim1.....=.qim+.nVt*.alpha_DD;
4953 .....
4954 .....if.(alpha1==.0).begin
4955 .....alpha1=.1e-5;
4956 .....end
x. 4957 .....H_fact.....=(qim1/.alpha1)*.(Dmob_dL/.Dvsat);

5463 .....if.(qim1==.0).begin
5464 .....Fj.....=.0.0;
5465 .....end.else.begin
5466 .....Fj.....=.0.5*(dps/.H_fact);
xi. 5467 .....end

```

BSIM-SOI 100.1.1 Beta2

Related to possible divide by 0 warnings: Following checks are added

```

if (NDEPEDGE_i <= 0.0) begin
    $error("Fatal: NDEPEDGE_i = %e is non-positive.", NDEPEDGE_i);
end

if (NGATE_i <= 0.0) begin
    $error("Fatal: NGATE_i = %e is non-positive.", NGATE_i);
end

if (XJ_i <= 0.0) begin
    $error("Fatal: XJ_i = %e is non-positive.", XJ_i);
end

if (NDIODE_i <= 0.0) begin
    $strobe("Warning: NDIODE_i = %e is non-positive, setting it to the default value.", NDIODE_i);
    NDIODE_i = 1.0;
end

if (NRECRO_i <= 0.0) begin

```

```

$strobe("Warning: NRECRO_i = %e is non-positive, setting it to the default value.", NRECRO_i);
NRECRO_i = 10.0;
end

if (NRECFO_i <= 0.0) begin
    $strobe("Warning: NRECFO_i = %e is non-positive, setting it to the default value.", NRECFO_i);
    NRECFO_i = 2.0;
end

```

Related to bias-dependent if-else conditions:

```

4350 ..... Ps ..... = (vgfb1-x7_s) * (vgfb1-x7_s) * (1.0/gam2) -. Ds;
4351 ..... Ps ..... = hypsmooth(Ps -. 1.0e-3, 1.0e-5) .+ 1.0e-3;
i. 4352 ..... sqms ..... = sqrt(Ps);

ii. 4367 ..... T2 = .lexp(UCS_a * .ln(0.5 * (1.0 + (qis /. qbs)))));

4411 ..... Vdsat ..... = hypsmooth(Vdsat -. 1.0e-3, 1.0e-5) .+ 1.0e-3;
4412 ..... Vdsat ..... = Vdsat /. AbulkIV;
iii. 4413 ..... T7 ..... = pow((Vds /. Vdsat) .+ 1.0e-6, 1.0 /. DELTA_t);

4711 ..... Pd ..... = (vgfb1-x7_d) * (vgfb1-x7_d) * (1.0/gam2) -. Dd;
4712 ..... Pd ..... = hypsmooth(Pd -. 1.0e-3, 1.0e-5) .+ 1.0e-3;
iv. 4713 ..... sqd ..... = sqrt(Pd);

v. 4717 ..... qid ..... = gam2 * Dd * nVt /. (xgd + gam * sqd);

4726 ..... Pm ..... = (vgfb1-x_m) * (vgfb1-x_m) * (1.0/gam2) -. Dm;
4727 ..... xgm ..... = gam * sqrt(Dm + Pm);
4728 ..... Pm ..... = hypsmooth(Pm -. 1.0e-3, 1.0e-5) .+ 1.0e-3;
vi. 4729 ..... sqm ..... = sqrt(Pm);

4747 ..... Pm ..... = (.vgfb1-x_m+.u_pd) * (.vgfb1-x_m+.u_pd) * (1/gam2) -. Dm /. km;
4748 ..... xgm ..... = gam * sqrt(Dm + Pm);
4749 ..... km0 ..... = 1.0 -. Em+.2.0 * (xgm * eta_p * inv_gam2);
4750 ..... x_ds ..... = x_ds * km * (d0 + D_bar) /. (km0 + km * D_bar);
4751 ..... dps ..... = x_ds * nVt;
4752 ..... Pm ..... = hypsmooth(Pm -. 1.0e-3, 1.0e-5) .+ 1.0e-3;
vii. 4753 ..... sqm ..... = sqrt(Pm);

4756 ..... if (abs(dps) > .1e-35) .begin
4757 ..... alpha_DD ..... = (qis -. qid) /. dps;
viii. 4758 ..... end

ix. 4766 ..... T2 ..... = .lexp(UCS_a * .ln(0.5 * (1.0 + (qim /. qbm)))));

4937 ..... alpha1 ..... = alpha_DD * (1.0 + 0.5 * (zsat * T0 * T0));
4938 ..... alpha1 ..... = hypsmooth(alpha1 -. 1.0e-3, 1.0e-5) .+ 1.0e-3;
4939 ..... qim1 ..... = qim + nVt * alpha_DD;
x. 4940 ..... H_fact ..... = (qim1 /. alpha1) * (Dmob_dL /. Dvsat);

xi. 5446 ..... Fj ..... = 0.5 * (dps /. H_fact);

```

N.B. – Similar changes have been made in EDGEFET and AGBCP2FET core calculations, as well.

16.2025bug2 (UCB/IITK): Updating the Technical Manual.

- The parameter table has been matched with the Verilog-A code.
- Minor updates have been made in the technical manual. For example, the name of the model has been corrected as BSIM-SOI 100.x.x in multiple places from BSIM6.BSIM-BULK/BSIM4.

17.2025bug3 (UCB/IITK): Removal of unused “tempc1” variable from code.

- The variable tempc1 was defined in the code but has never been used after assignment. Hence, it is now removed.
- The following update has been made in this regard:

BSIM-SOI 100.1.1 Beta1

```
4699 .....if.(tempc.>=0).begin
4700 .....tempc1.....=pC+.sqrt(tempc);
4701 .....end.else.begin
4702 .....tempc1=.0.0;
4703 .....end
4704 .....if.(tempc1.<=.0.0).begin
4705 .....x_ds.....=.0.0;
4706 .....end.else.begin
4707 .....x_ds.....=.2.0.*(qC./(pC+.sqrt(tempc)));
4708 .....end
```

BSIM-SOI 100.1.1 Beta2

```
4701 .....if.(tempc.>=0.0).begin
4702 .....x_ds.....=.2.0.*(qC./(pC+.sqrt(tempc)));
4703 .....end
```

N.B. – Similar changes have been made in EDGEFET and AGBCP2FET core calculations, as well.

----- End of Document-----